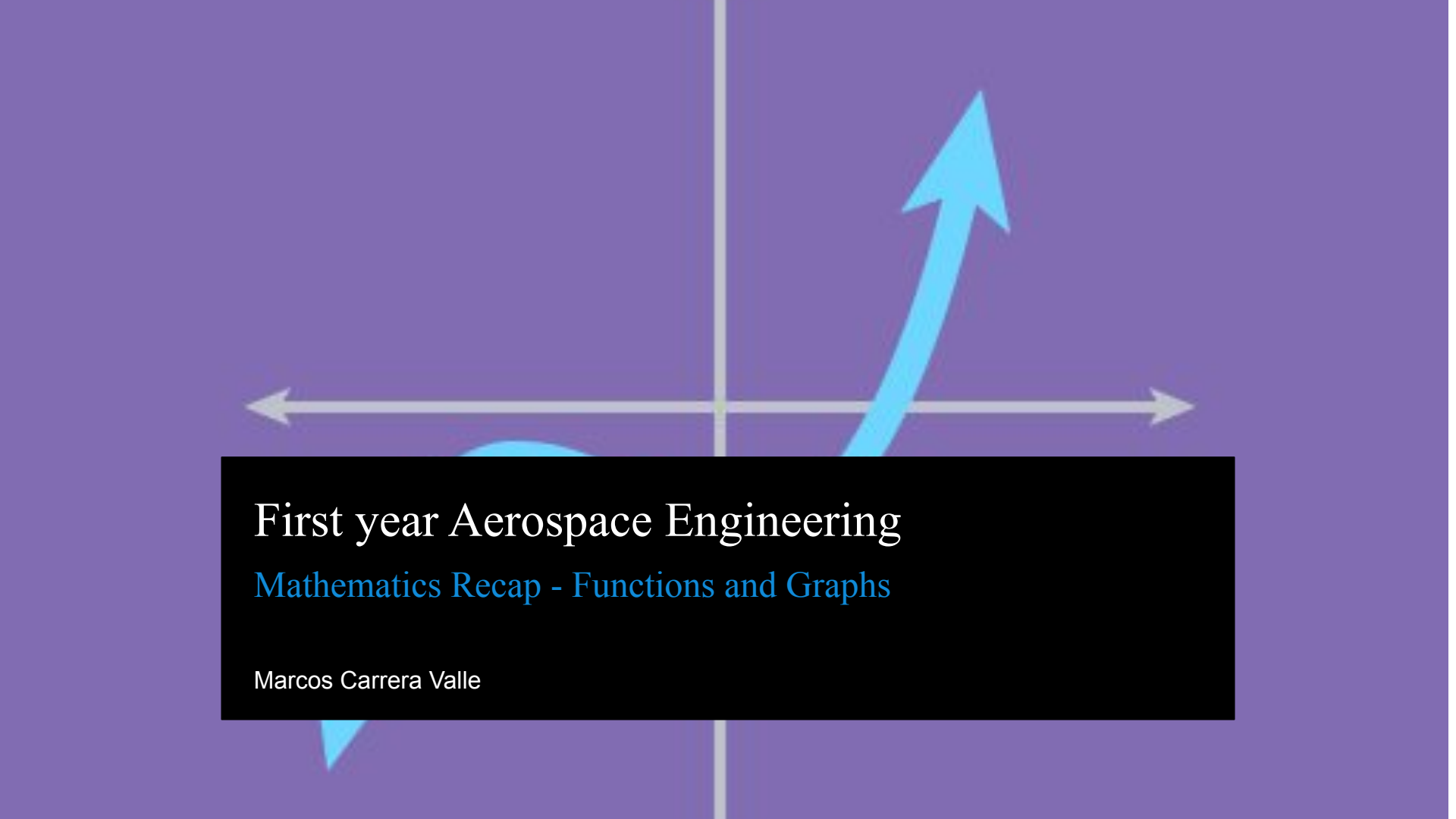




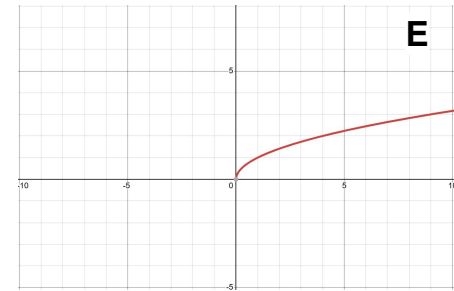
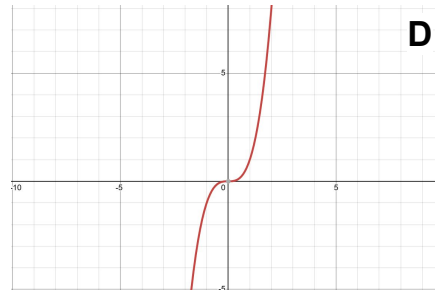
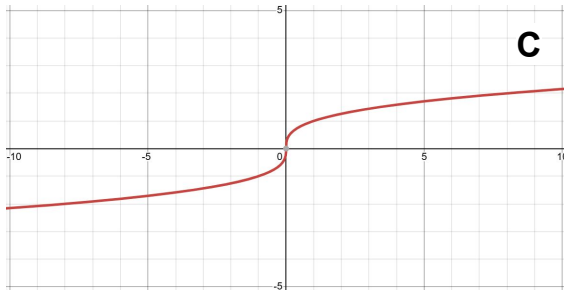
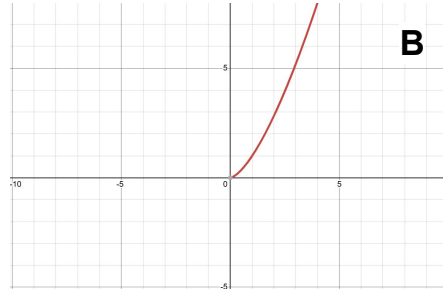
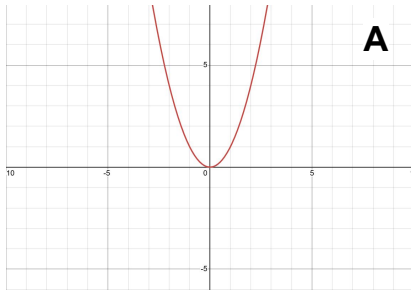
First year Aerospace Engineering

Mathematics Recap - Functions and Graphs

Marcos Carrera Valle

Recognize functions - 5 points

Match each graph to its function

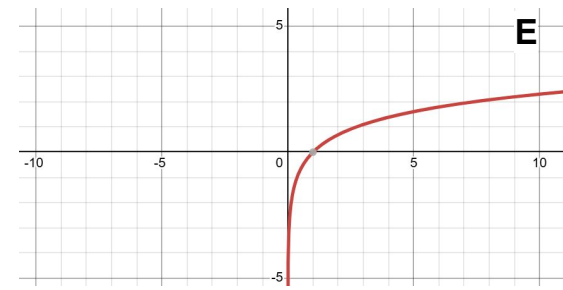
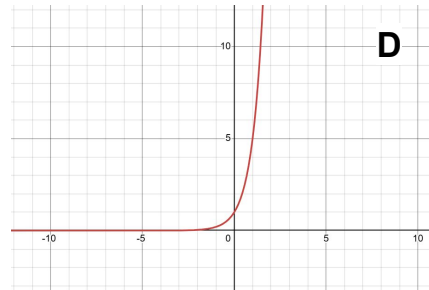
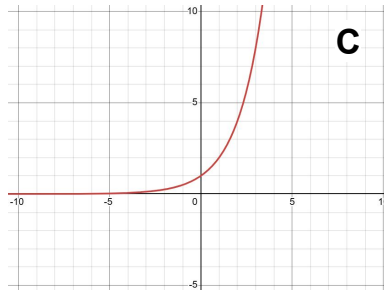
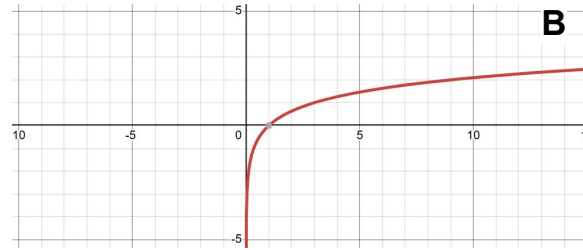
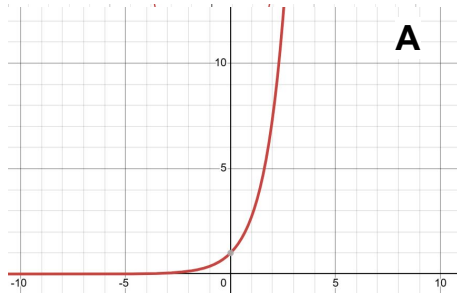


- 1) x^2
- 2) x^3
- 3) $\sqrt{x}$
- 4) $\sqrt[3]{x}$
- 5) $\sqrt[3]{x^2}$

Recognize functions - 5 points

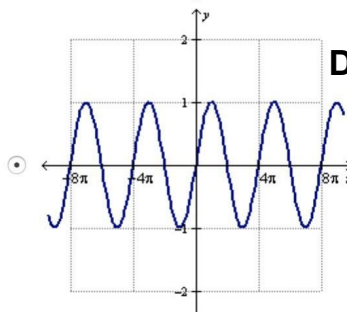
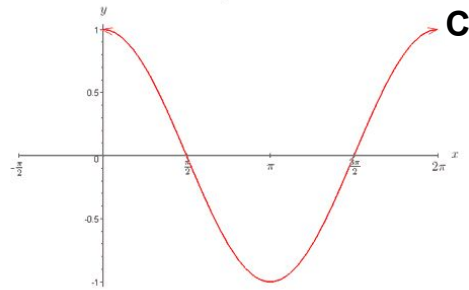
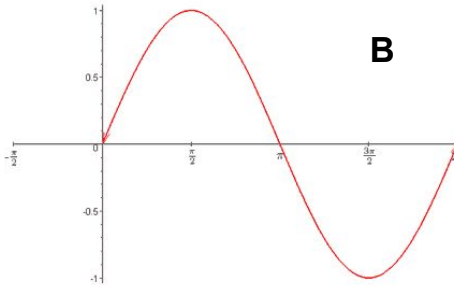
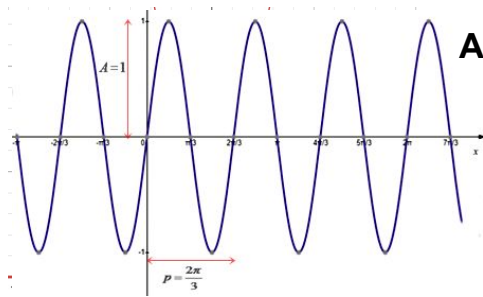
Match each graph to its function

- 1) 2^x
- 2) 5^x
- 3) $\log_3 x$
- 4) $\exp^x$
- 5) $\ln x$



Recognize functions - 4 points

Match each graph to its function



- 1) $\sin x$
- 2) $\cos x$
- 3) $\sin 3x$
- 4) $\sin \frac{x}{2}$

Sketch functions, domain, range, asymptote

- 1) Sketch the graph of the function $f(x) = 3\cos(x) + 1$, $-2\pi < x < 2\pi$ - 2 points
- 2) What is the domain of $f(x)$ - 1 point
- 3) What is the range - 1 point
- 4) Are there any asymptotes? If so, give the equations of the asymptote lines - 1 point

Answers

Slide 2: 1A, 2D, 3E, 4C, 5B

Slide 3: 1C, 2D, 3B, 4A, 5E

Slide 4: 1B, 2C, 3A, 4D

Slide 5:

Domain = $\mathbb{R}$ (All real numbers)

Range = $2 < f(x) < 4$

Asymptotes = none

